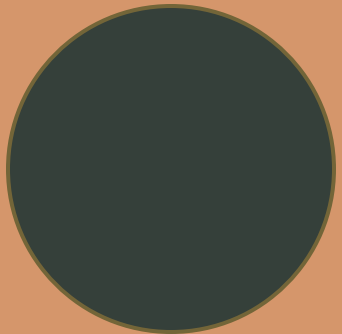




Covid-19 Model Prediction using Neural networks

Intended Audience: Medical
Professionals

By: Poorvi Raut



AGENDA

- 1.PROJECT DEFINITION
2. DATASET OVERVIEW
3. DATA CLEANING and PRE-PROCESSING
4. MACHINE LEARNING MODELS
5. CONCLUSION AND FUTURE ENHANCEMENTS

PROJECT DEFINITION

- The project aims to determine whether a **COVID-19 patient will survive or not** based on key health indicators. Predicting survival requires analyzing multiple factors, including **age, oxygen levels, severity, and underlying conditions**.
- COVID-19 poses a serious threat to **public health**, and accurate predictions can help hospitals allocate resources effectively. The goal is to develop a system that can **precisely classify patients** based on survival likelihood, aiding in better medical decision-making.



DATASET OVERVIEW

Reference: [Titanic DataSet Kaggle](#)

This dataset is structured similarly to the **Titanic survival prediction dataset**, where the goal is to classify whether a COVID-19 patient **survives (1) or does not survive (0)** based on medical and demographic features.

- **Each row** represents a **single patient case**
- **Each column** represents a **key attribute** such as **Age, Gender, Oxygen Level, Severity, and Treatment Factors**



DATASET OVERVIEW CONTINUED:

- **final_train.csv**
- **Contains** training data for model development
- **Rows:891**
- **Columns: 12** features including **Survived (Target Variable)**
- **final_test.csv**
- **Used for evaluating model performance**
- **Rows: 418**
- **Columns: Same as final_train but without Survived so 11**
- **final_gender_submission.csv**
- **Contains sample predictions for final_test patients**
- **Rows: 418 (Same as final_test)**
- **Columns: 2 Patient ID, Predicted Survival (0/1)**

DATASET FEATURES

Features	Description	final_train.csv	final_test.csv	final_gender_submission.csv
PatientID	Unique identifier for each COVID-19 patient.	Yes	Yes	Yes
Severity Level	Indicates the degree of illness experienced by the COVID-19 patient (e.g., mild -1, moderate-2, severe-3).	Yes	Yes	No
FullName	The complete name of the individual diagnosed with COVID-19.	Yes	Yes	No
Gender	The biological sex of the COVID-19 patient, Male/Female	Yes	Yes	No
Age	The patient's age	Yes	Yes	No
CloseContacts	Number of individuals who had close contact with the COVID-19 patient	Yes	Yes	No
FamilyInfected	Indicates if other family members of the patient were also infected with COVID-19	Yes	Yes	No
Case ID	A specific identifier for the COVID-19 case	Yes	Yes	No
TreatmentCost	The expenses associated with the medical care provided to the COVID-19 patient.	Yes	Yes	No
IsolationWard	The specific ward where the COVID-19 patient was isolated to prevent further spread.	Yes	Yes	No
HospitalLocation	Hospital where the patient was admitted	Yes	Yes	No
Survived	Patient survival status (1/0)	Yes	No	Yes

3. DATA CLEANING AND PREPROCESSING

Data cleaning and preprocessing is a crucial step in Data Science for model prediction. It involves identifying and correcting errors in the dataset by handling missing values, removing duplicates, and fixing inconsistencies. It ensures the data is accurate, consistent, and ready for analysis or machine learning.

Steps Involved include:

- 1. Handling missing values and Feature encoding**
- 2. Feature Engineering**
- 3. Exploratory Data Analysis**

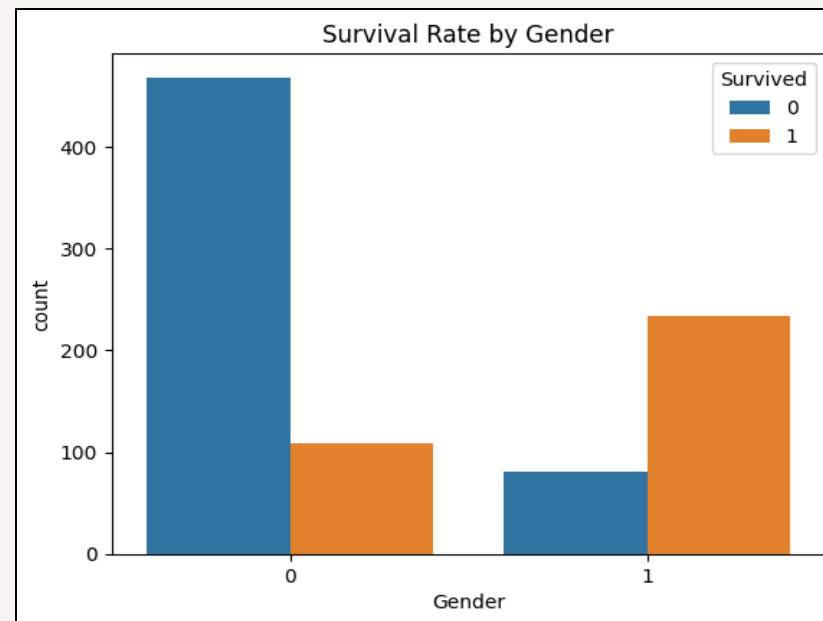
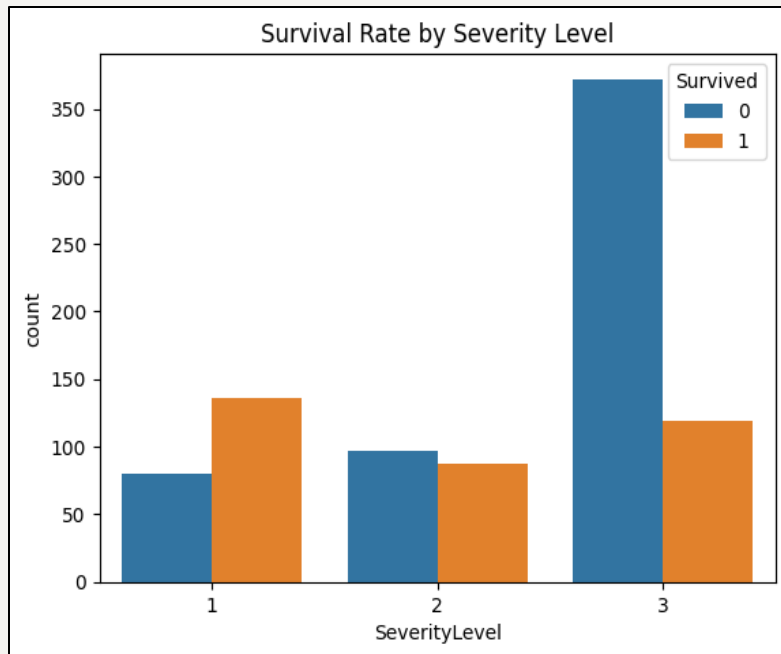


3.1 HANDLING MISSING VALUES

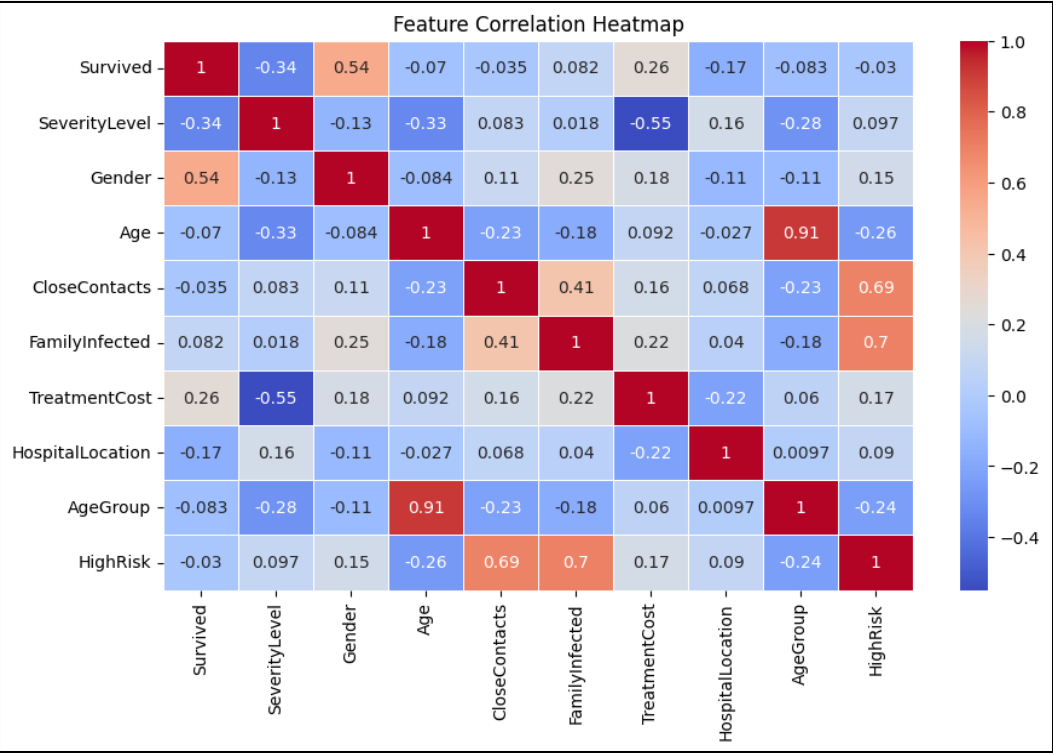
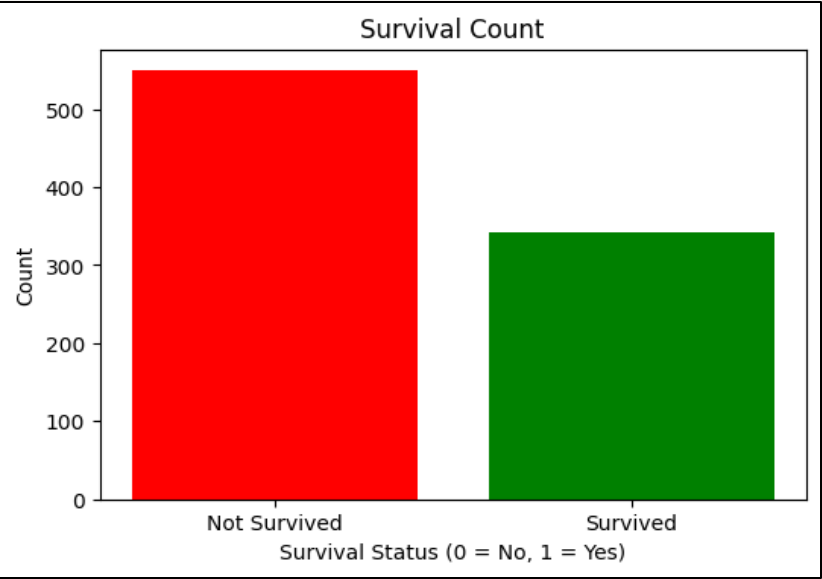
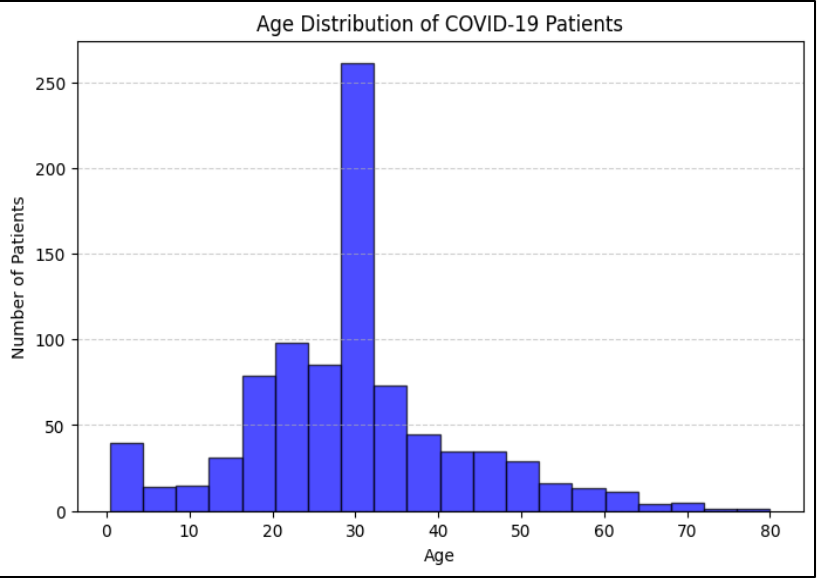
- Under **Data Cleaning & Preprocessing**, the **Handling Missing Values** step ensures data completeness by addressing gaps in the dataset.
- For our given **final_train.csv** and **final_test.csv** files
 - checked for any missing values as first step. `final_train.isnull().values.any()`
 - We filled all numerical numerical with **mean** and all categorical columns with **mode**
 - Converted Categorical variables to numeric data like **Gender** (Male/Female → 0/1) and **HospitalLocation** {'C': 0, 'Q': 1, 'S': 2}
 - Finally Dropped few irrelevant columns from **final_train.csv** and **final_test.csv** files like 'PatientID', 'FullName', 'CaseID', 'IsolationWard'

3.2 EXPLORATORY DATA ANALYSIS

- Now, will explore every features, so we can remove unnecessary data-columns and select only those data-columns which has significance to classify the target variable which is **Survived** in our case with **Binary value 0 :Not Survived and 1:Survived** and plot visualizations.



FURTHER VISUALIZATIONS



3.3 FEATURE ENGINEERING

- Create new features based on existing ones.
- Age Group: Categorize age into groups.
- High Risk: Identify patients with more CloseContacts and FamilyInfected.

	Survived	SeverityLevel	Gender	Age	CloseContacts	FamilyInfected	\
0	0	3	0	22.0	1	0	
1	1	1	1	38.0	1	0	
2	1	3	1	26.0	0	0	
3	1	1	1	35.0	1	0	
4	0	3	0	35.0	0	0	
	TreatmentCost	HospitalLocation	AgeGroup	HighRisk			
0	7.2500		2	1	0		
1	71.2833		0	1	0		
2	7.9250		2	1	0		
3	53.1000		2	1	0		
4	8.0500		2	1	0		

4. ML MODELS

- Following data preparation (overview, cleaning, preprocessing), the next key step is training an ML model for unseen data prediction.
We employed Neural nets : Shallow ANN and ANN with Back propagation for model prediction.

Neural Network Models in Machine Learning



1. Shallow ANN

- A shallow Artificial Neural Network (ANN) is a neural network with **only one hidden layer** between the input and output layers.
- **Small Equation (single hidden neuron):**

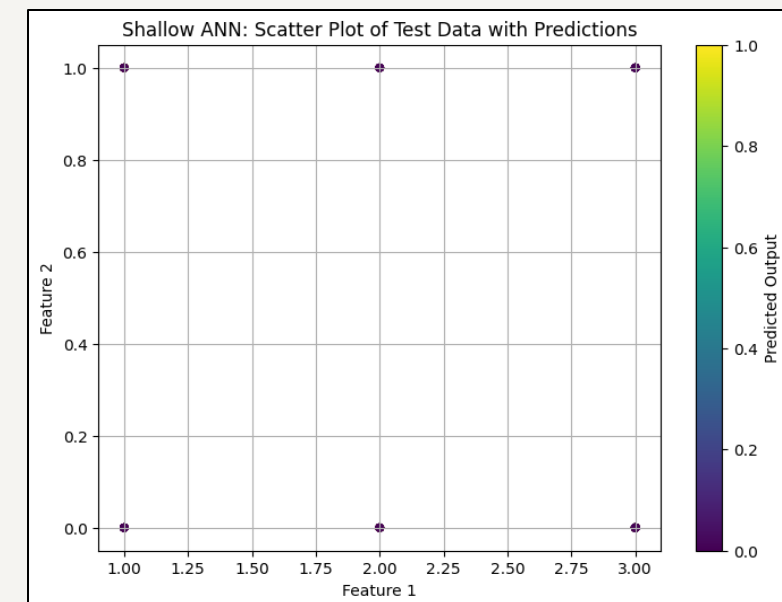
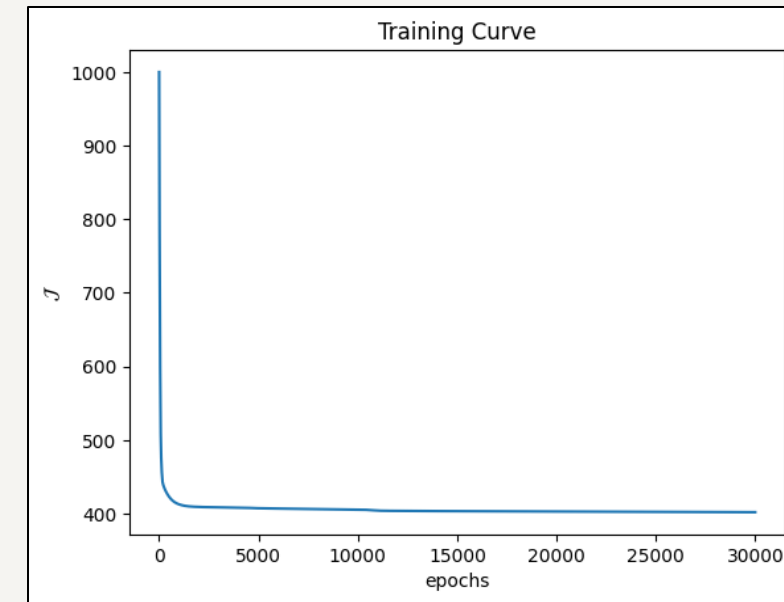
$$a(1)=f(\sum wx+b)$$

- We trained the model by fine tuning hyper-parameters eta and epsilon and got an accuracy of 72%

```

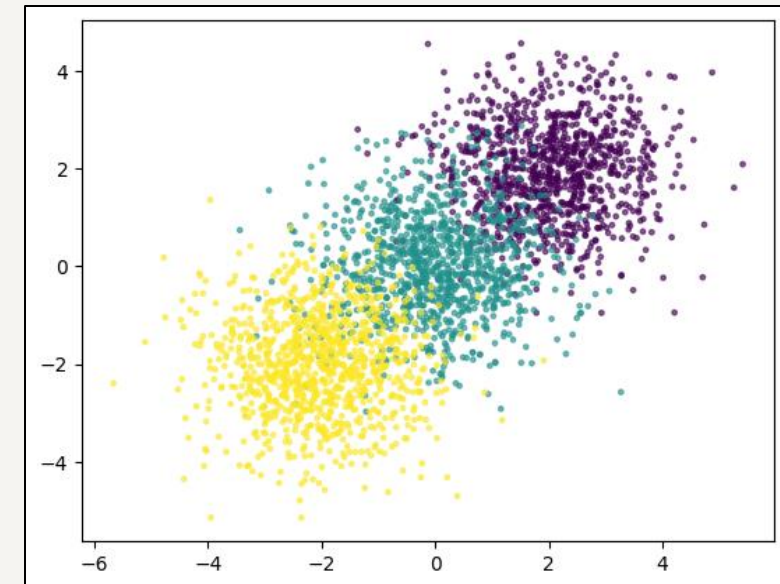
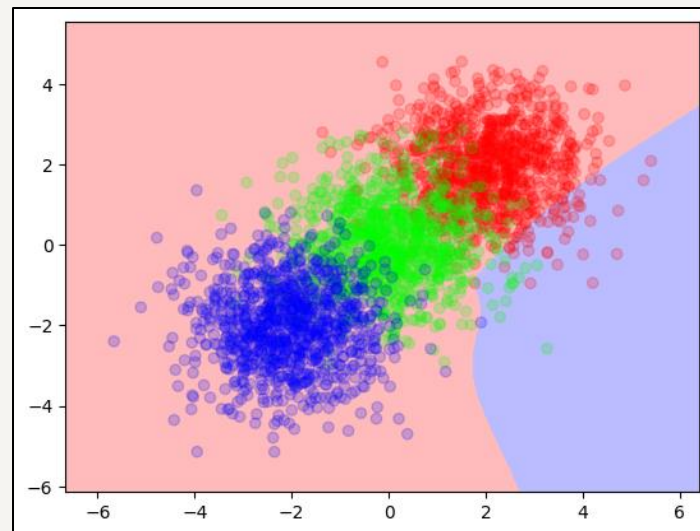
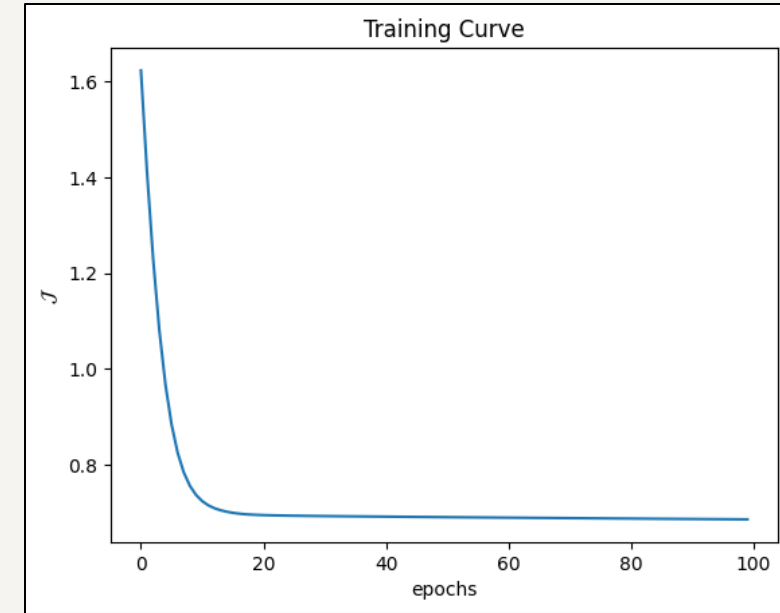
Accuracy: 0.7201492537313433
{1: array([[ -0.24458904, -0.7456028 , -1.03360539],
 [ 0.34820883, -0.68768087, -1.09458612],
 [ 0.31735376,  1.44142309, -0.27001826],
 [ 0.90026296,  1.58453578, -0.44565397],
 [-0.12036237,  0.75264963, -1.02946292],
 [ 0.00731396, -0.40780102, -1.45890332],
 [-0.6287236 ,  0.02238768, -0.9166253 ],
 [ 0.66762508,  0.10067679,  0.05267754],
 [ 0.95825752, -0.26709085,  0.99973493]]), 2: array([[ 0.43883312, -1.69472202],
 [ 0.44903817,  0.02519997],
 [-0.57374084, -0.17713903]])}
{1: array([ -1.0678747 , -0.54912703, -0.79291074]), 2: array([ 0.06542135,  2.43255014])}

```



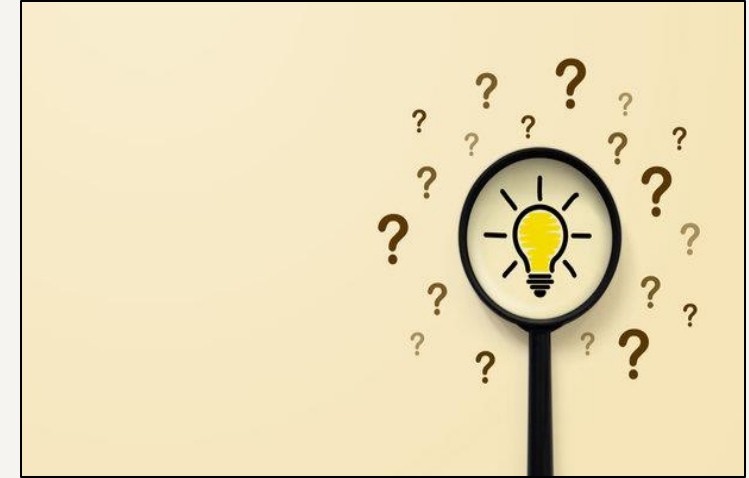
2. ANN with Variable Architecture and Back Propagation

- An Artificial Neural Network (ANN) with **variable architecture** can have a flexible number of layers and neurons. It learns through **backpropagation**, adjusting weights based on the error between predictions and actual values.
- We trained the model by fine tuning hyper-parameters eta and epsilon and got an Training accuracy of 0.6940



5. CONCLUSION AND FUTURE ENHANCEMENTS

- The simple, one-layer network (shallow ANN) did a bit better (72%) at predicting than the more complex network that learns by fixing its mistakes (backpropagation ANN at 69%).
- For future enhancements we can fine tune the hyperparameters, improve existing features and combine models (Ensemble Learning) for better results



APPENDIX

- The visuals are sourced from Google Search, excel ,google sheets and the code was created using the Google Colab IDE.
- Dataset Source : [Titanic DataSet Kaggle](#)

Thank you